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5. (a) Discuss the role of any two imaging techniques in the diagnosis and monitoring of brain disorders, highlighting their clinical significance. 9
- (b) Explain the principle, working and clinical significance of EEG. 6
6. Write short notes on any **three** of the following. 15
- (a) Ethical issues in medical diagnostics
- (b) WIDAL test
- (c) Personalized medicine
- (d) Immunization programs for pregnant women

07/06/2026 (MORNING)

[This question paper contains 4 printed pages]

Your Roll No.

: 07.06.2026 (M)

Sl. No. of Q. Paper

: 8516

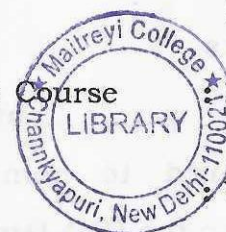
Unique Paper Code

: 2233520002 2233520005

Name of the Paper

: DSE-Medical Diagnostics
and Healthcare NEP -
UGCF

Name of the Course



: B.Sc. (H) Zoology/B.Sc.
(P) Life Sciences

Semester

VIII

Time : 2 Hours

Maximum Marks : 60

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Attempt any **four** questions in **all**. Question **No.1** which is compulsory.
1. (a) Define the following. 5
- (i) Agglutination
- (ii) Homeostasis
- (iii) Atherosclerosis
- (iv) Hemogram
- (v) Hematuria

- (b) Differentiate between the following. 6
- (i) Serum and Plasma
 - (ii) Endoscopy and colonoscopy
 - (iii) High-Density Lipoprotein and Very Low-Density Lipoprotein
- (c) Fill in the blanks. 4
- (i) During the step of PCR, primers bind to complementary sequences on the DNA template.
 - (ii) Increased levels of in blood indicate impaired kidney function and reduced filtration efficiency.
 - (iii) The antigen is associated with the flagellar component of Salmonella.
 - (iv) The National Immunization Program aims to protect against diseases.

2. (a) Discuss healthcare screening programs for Diabetes and cancer, with their advantages and limitations. 9
- (b) Outline the causes, diagnosis, and prevention of cardiovascular diseases. 6
3. (a) Explain the Liver Function Test (LFT) and Kidney Function Test (KFT) in detail, including parameters and their clinical interpretation. 9
- (b) Evaluate the significance of lipid profile test in disease diagnosis, with reference to cholesterol and triglycerides. 6
4. (a) Schematically represent the steps involved in Polymerase Chain Reaction (PCR) technique. Highlight its components and applications in molecular diagnostics. 9
- (b) Discuss the significance of ELISA in clinical diagnosis of infectious diseases. 6